

SIH WINNING PROJECT – CASE STUDY ANALYSIS PROFORMA

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CASE STUDY – 1

Item	Details
1. Project Title	Smart Sabarmati Riverfront & Urban Water Bodies
2. SIH Year	2025
3. Winning Institution	NIT, Rourkela
4. Team Name	SWADES
5. Problem Statement	Develop smart, technology-driven solutions for the Sabarmati Riverfront and other urban water bodies, addressing issues such as water quality degradation, waste management, and public safety.
6. Domain	Smart Automation, Smart Cities and Environmental Sustainability
7. Software / Hardware	Both Software and Hardware — autonomous river-surface cleaning robot, sensors/intelligent systems, and the Sabarmati Mitra mobile application.

A. Understand the Problem

What real-world problem does the project solve?

The project solves the problem of pollution, waste accumulation, and safety issues in the Sabarmati Riverfront and other urban water bodies. The main goal was to make these water bodies cleaner and safer by using smart technology

Who is affected by this problem?

The problem affects people living near the Sabarmati Riverfront and other urban water bodies. Floating waste, water pollution and safety risks can affect the public, cleaning workers and the surrounding environment.

Why is solving this problem important?

Solving this problem is important to keep the urban water bodies clean, safe and environmentally healthy, also by reducing waste and pollution can improve water quality and protect aquatic life while smart technology and safety systems can help reduce risks to people.

B. Understand the Solution

What is the proposed solution?

The proposed solution by the team was an autonomous river-surface cleaning robot combined with a smart monitoring and alert system. The robot removes floating waste from the water surface, while the system can detect drowning incidents and provide alerts through the Sabarmati Mitra mobile application. It also provides real-time water-quality information to help monitor the condition of the water body.

Draw a simple block diagram of the solution:

Problem → Input / Data → Processing → Solution → Output / Result

PROBLEM

Pollution, floating waste & safety issues

INPUT / DATA

Floating waste + Water-quality data + Sensor data + Safety information

PROCESSING

Sensors + AI/Smart monitoring + Autonomous robotic system

SOLUTION

Autonomous river-surface cleaning + Smart monitoring & safety system

OUTPUT / RESULT

Cleaner water bodies + Water-quality information + Faster safety alerts

What technologies are used?

Technologies used are Programming, Artificial Intelligence, IoT , Mobile Application, Sensors, Other: Robotics and Autonomous system

C. Innovation

What is new or different about this project?

The project combines autonomous river-surface cleaning with public-safety and water-quality monitoring in one integrated system. The robot can remove floating waste, while the system can detect drowning incidents, send alerts through the Sabarmati Mitra mobile application, and provide real-time water-quality information. This combination makes the solution more comprehensive than focusing only on cleaning the water surface

What existing method does it improve or replace?

It improves the approach of manual monitoring and removal of floating waste from urban water bodies by introducing an autonomous robotic system. It also improves safety monitoring by using technology to detect drowning incidents and send alerts, reducing dependence on continuous manual observation

D. Impact

Who will benefit from this project?

The project will benefit people visiting the Sabarmati Riverfront, communities living near urban water bodies, municipal authorities, cleaning workers, and the surrounding aquatic environment. It can help authorities manage water bodies more effectively while improving public safety.

What is the expected benefit?

The expected benefits are cleaner urban water bodies, better monitoring of water quality, reduced human effort in removing floating waste, and improved public safety through faster detection and alerts. The project therefore supports more efficient and sustainable management of urban water bodies.

It Saves time, Improves safety, Reduces human effort, Improves accuracy, Protects environment,

Other: Better water body monitoring

CASE STUDY – 2

Item	Student Response
Project Title	Fake Social Media Account Detection
SIH Year	2024
Winning Institution	G H Rasoni College of Engineering and Management, Pune
Team Name	Tech Simp-2
Problem Statement	Detect fake social media accounts and identify suspicious profiles that may be created for fraud, impersonation or other harmful activities.
Domain	Artificial Intelligence, Machine Learning , Cybersecurity
Target Users	Social media users, platforms, organizations and cybersecurity teams
Proposed Solution	Develop an AI/ML-based system that analyzes different characteristics of a social media profile and predicts whether the account is genuine or fake.
Technologies Used	Artificial Intelligence, Machine Learning, data analysis and software/web technologies
Input / Data	Profile information such as account activity, follower/following details, profile characteristics and other available account features
Processing	The collected features are analyzed using machine learning techniques to identify patterns associated with fake accounts.
Output	The system predicts whether an account is likely to be genuine or fake/suspicious.
Innovation	Uses AI/ML to automatically identify suspicious social media accounts instead of depending only on manual reporting.
Social / Industry Impact	Can help reduce impersonation, scams, fake profiles and other forms of online misuse, making social media safer.
What did I learn?	I learned how AI and machine learning can be applied to cybersecurity problems and how real-world data can be processed to detect suspicious activities.

COMPARISON OF TWO WINNING PROJECTS

Parameter	Case Study 1	Case Study 2
Real-world problem	Pollution, floating waste, water-quality issues and public-safety problems in urban water bodies.	Fake social media accounts, impersonation, scams and online misuse.
Target users	Public, cleaning workers, municipal authorities and communities near water bodies.	Social media users, social media platforms, organizations and cybersecurity teams.

Main technology	Robotics, IoT, sensors, AI and mobile application.	Artificial Intelligence, Machine Learning, Python and data analysis.
Input / data	Floating waste, water-quality data, sensor data and safety information.	Profile information, account activity and other social-media account features.
Processing	Sensors collect information; AI/smart monitoring analyzes it and an autonomous robot removes floating waste.	Machine-learning techniques analyze account features and identify patterns associated with fake accounts.
Output	Cleaner water bodies, water-quality information and faster safety alerts.	Prediction of whether an account is genuine or fake/suspicious.
Innovation	Combines autonomous cleaning, water-quality monitoring and public-safety systems in one solution.	Uses AI/ML to automatically identify suspicious social-media accounts.
Social impact	Reduces pollution, protects the environment, improves public safety and reduces human effort.	Helps reduce fake accounts, impersonation and online scams and improves digital safety.
Difficulty level	High – requires robotics, sensors, IoT, AI and hardware integration.	Medium and High – requires machine learning, data collection and software development.
Can students build a basic prototype?	Yes, but a simplified prototype would be easier using a small robot, sensors and basic IoT.	Yes, a basic AI/ML model and web application can be developed using publicly available datasets.

STUDENT ANALYSIS

Which project do you like better?

I like Case Study 2 – Fake Social Media Account Detection better because it solves a common problem faced by social media users and combines artificial intelligence with cybersecurity.

Why did you select it?

I selected it because fake accounts can be used for impersonation, scams and spreading misleading information. I am also interested in computer science and cybersecurity, so this project is useful and relevant to my interests.

What part of the project impressed you most?

The part that impressed me most is the use of machine learning to analyze account characteristics and automatically identify suspicious or fake profiles. This can reduce the need for completely manual detection

What technology did you learn from this project?

I learned about Artificial Intelligence, Machine Learning, Python and data analysis, and how these technologies can be used to solve practical cybersecurity problems.

Can you improve this project? If yes, what improvement would you make?

Yes. I would improve it by adding real-time detection and a simple browser or mobile interface. I would also design an AI system that can analyze a social media post or profile picture (PFP) and detect whether it is genuine or potentially fake/AI-generated. The system could provide a confidence score and explain the characteristics that made the content suspicious

MY OWN MINI SIH IDEA

Problem I would like to solve	Students often forget assignments, deadlines and important academic tasks.
Who has this problem?	School and college students.
My proposed solution	A simple smart student task manager that stores assignments and deadlines and reminds students about upcoming tasks.
Technology I would use	HTML, CSS, JavaScript, Python and a simple database.
Expected benefit	Students can organize their academic work, avoid missing deadlines and manage their time better.
Simple sketch / block diagram	Student enters task → System stores and processes deadline → Reminder/notification → Student completes task

Suggested format: INPUT → PROCESSING → OUTPUT

FACULTY EVALUATION RUBRIC – 20 MARKS

Criteria	Marks
Understanding of problem	4
Understanding of solution	4
Technology identification	3
Comparison of projects	2
Innovation proposed	4
Presentation	3
TOTAL	20

LEARNING OUTCOME

After completing the activity, the student should be able to: Identify a real-world engineering problem, understand how technology can address it, analyze an existing successful solution, and propose a simple technology-based solution.

SOURCE

Case study 1:

<https://indiaeducationdiary.in/nit-rourkela-students-emerge-national-winners-at-smart-india-hackathon-2025/>

Case study 2:

<https://www.punekarnews.in/ghrcem-students-win-first-prize-at-smart-india-hackathon-2024/>

